

Cheat sheet: graphical Rasch models

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Introduction

A **graphical Rasch model** is an extension of the Rasch model that can incorporate differential item functioning (**DIF**) and local dependence (**LD**).

Notation

Let $X = (X_1, \dots, X_k)$ denote the vector of item responses, let $Z = (Z_1, \dots, Z_m)$ denote a vector of covariates (potential DIF variables), and let θ denote the latent variable that the item measure.

Rasch model

The Rasch model for polytomous items assumes local independence

$$Prob(X = x|\theta) = \prod_{i=1}^k Prob(X_i = x_i|\theta) \quad (1)$$

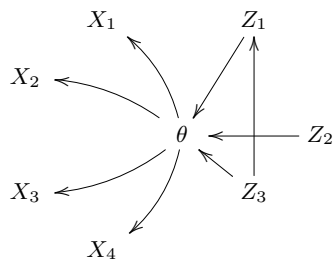
and that the parametric form is

$$Prob(X_i = x_i|\theta, Z) = \exp(\theta x_i + B(i, x_i))/K_i(\theta, B) \quad (2)$$

where B is a matrix of item responses and K is a normalizing that does not depend on x . Note that the fact that left side of equation (2) does not depend on Z . This is due to the assumption of no **DIF**. Combining (1) and (2) yields

$$Prob(X = x|\theta, Z) = \exp(\theta S + \sum_i B_i(i, x_i))/K \quad (3)$$

Where $S = \sum_i x_i$ is the sum score and $K = \sum_i K_i(\theta, B_i)$.



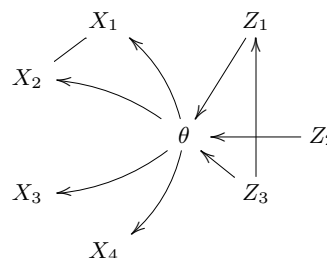
The graphical Rasch model incorporates **LD** and **DIF** by expanding the log-linear structure of the probabilities in equation (3).

Adding LD

LD between, say, items X_1 and X_2 can be included by fitting the model

$$Prob(X = x|\theta, Z) = \exp(\theta S + \sum_i B_i(i, x_i) + D(x_1, x_2))/K \quad (4)$$

where D is a matrix of interaction parameters. The models (3) and (4) are nested and testing (4) against (3) using conditional likelihood ratio test yields a test of **LD**.



Testlets (super-items)

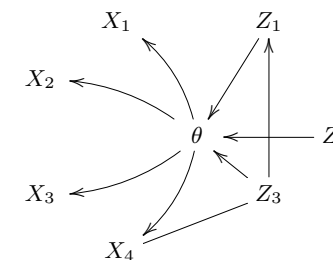
Combining the item pairs with **LD** into single combination items provides a straightforward way of modelling **LD**. For two items, say, X_1 and X_2 this can be done by replacing them with the a item $X_{1,2} = X_1 + X_2$.

Adding DIF

DIF for, say, item X_4 with respect to the covariate Z_3 can be included by fitting the model

$$Prob(X = x|\theta, Z) = \exp(\theta S + \sum_i B_i(i, x_i) + L(x_4, z_3))/K \quad (5)$$

where L is a matrix of interaction parameters. The models (3) and (5) are nested and testing (5) against (3) using conditional likelihood ratio test yields a test of **DIF**.



Splitting an item for DIF

Splitting an item provides a straightforward way of modelling DIF.

References

- [1] S. Kreiner and K. B. Christensen, "Graphical rasch models," pp. 187–203, 2002. [Online]. Available: http://link.springer.com/10.1007/978-1-4757-3625-0_15
- [2] —, "Analysis of local dependence and multidimensionality in graphical loglinear rasch models," *Communications in Statistics - Theory and Methods*, vol. 33, pp. 1239–1276, 12 2004. [Online]. Available: <http://www.tandfonline.com/doi/abs/10.1081/STA-120030148>
- [3] —, "Validity and objectivity in health-related scales: Analysis by graphical loglinear rasch models," pp. 329–346, 2007. [Online]. Available: http://link.springer.com/10.1007/978-0-387-49839-3_21
- [4] —, "Item screening in graphical loglinear rasch models," *Psychometrika*, vol. 76, pp. 228–256, 4 2011. [Online]. Available: <http://link.springer.com/10.1007/s11336-011-9203-y>
- [5] S. Kreiner and T. Nielsen, *Item analysis in DIGRAM 3.04: Part I: Guided tours*. Department of Biostatistics, University of Copenhagen, 2013. [Online]. Available: <https://biostat.ku.dk/DIGRAM/Item%20analysis%20in%20DIGRAM%205-01%20-%20guided%20tours.pdf>